Semisupervised SAR Ship Detection Network via Scene Characteristic Learning

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***Abstract*— In recent years, target detection methods based on deep learning have achieved extensive development in synthetic aperture radar (SAR) ship detection. However, training such detectors requires target-level annotations of SAR images that are hard to be obtained in practice. To reduce the dependence of network training on expensive target-level annotations, we propose a novel semisupervised SAR ship detection network via scene characteristic learning. The proposed network focuses on utilizing the scene-level annotations of SAR images to improve the detection performance in the case of limited target-level annotations. Compared with the traditional fully supervised SAR ship detection network, the proposed network constructs a scene characteristic learning branch parallel with the detection branch. In the scene characteristic learning branch, a scene classification loss and a scene aggregation loss are designed to utilize the scene-level annotations. Under the constraint of these two losses, the feature extraction network can fully learn the scene characteristics of SAR images, thus enhancing its feature representation ability for ship targets and clutter. In addition, we propose a hierarchical test process from scene to target. After recognizing the scene types of input SAR images, we design different detection strategies for SAR images recognized as different scenes. The proposed test process can significantly reduce the inland and inshore false alarms, thus leading to higher detection performance. The experiments based on two measured SAR ship detection datasets demonstrate the effectiveness of the proposed method.**

***Index Terms*— Hierarchical test process from scene to target, scene characteristic learning, semisupervised learning, ship detection, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) has unique advantages in that it is not limited by time, illumination, and weather constraints. At present, with the development of SAR sensor technology, a large area of high-resolution SAR images can be obtained. And SAR automatic target detection, the first step in SAR image interpretation [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], has become a vital issue that the researchers pay attention to. In this field, an important branch is SAR ship detection, which is of great significance in marine surveillance and military intelligence acquisition.

In the past few decades, many traditional SAR ship detection methods have been proposed, e.g., the detection methods based on statistical characteristics [1], [2], visual saliency [3], [4], and transformation [6]. The constant false alarm rate (CFAR) method [1] is based on statistical characteristics and has been widely studied. However, the performance of CFAR method is greatly affected by the statistical modeling of the background clutter. Some improved techniques [7], [8] of the CFAR method can accurately fit sea clutter, thus achieving a good performance in simple offshore scenes. However, their performance drops significantly for the complex inshore scenes.

With the improvement of computing power, the detection methods based on deep learning are developing rapidly. These methods can be roughly divided into two-stage detection methods (e.g., Faster R-CNN series [26], [27]) and one-stage detection methods (e.g., single-shot multibox detector (SSD) [28] and RefineDet [29]). Among them, RefineDet introduces the idea of two-step cascaded classification and regression into one-stage detector, achieving a tradeoff between speed and accuracy. Besides optical images, deep learning methods have also achieved a good performance in SAR target detection [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20]. Zhang et al. [9] proposed HyperLi-Net to achieve high-accuracy and high-speed SAR ship detection. Du et al. [13] proposed a saliency-guided SSD to mitigate the problem of clutter interference. Cui et al. [14] fused the high-level and low-level feature maps so that the fused feature maps contained richer information. Xu et al. [15] proposed a lightweight SAR ship detector called Lite-YOLOv5, which realizes the on-board ship detection without sacrificing the accuracy.

Although the SAR ship detection methods based on deep learning can improve the detection performance to a certain extent, they still face some problems. First, most of these methods rely heavily on the target-level annotations. Nevertheless, the target-level labeling process is extremely difficult. The self-supervised learning [30], [31] and semisupervised learning [32], [33], [34], [35], [36], [37] can alleviate this problem. Self-supervised learning can effectively utilize unlabeled images by designing suitable pretext tasks. However, it is essentially a subset of unsupervised learning methods, and it

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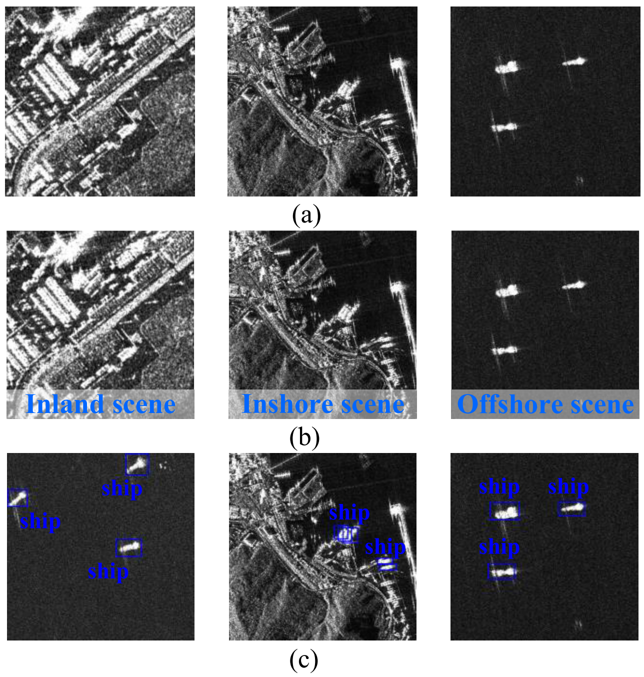


Fig. 1. SAR images labeled at different levels: (a) unlabeled SAR images; (b) scene-level labeled SAR images; and (c) target-level labeled SAR images.

is difficult to achieve satisfactory performance without using target-level annotations. Nevertheless, the detection methods based on semisupervised learning can simultaneously use a small amount of target-level labeled images and a large amount of unlabeled or scene-level labeled images, which not only reduce the dependence on expensive target-level annotations but also ensure a high detection performance. Therefore, this article focuses on the SAR ship detection method based on semisupervised learning. Fig. 1 shows the SAR images labeled at different levels used for ship detection. For the scene-level labeled SAR images, the scene type of each SAR image sample needs to be recognized manually. But for the target-level labeled SAR images, the specific positions of all ship targets in each SAR image sample need to be marked manually. Obviously, the scene-level labeled and unlabeled SAR images are easier to be obtained than the target-level labeled SAR images. In addition, the scene-level labeled SAR images contain richer annotation information than the unlabeled SAR images.

Several studies have applied semisupervised learning to target detection. Sohn et al. [33] proposed STAC, which assigned pseudo labels for unlabeled images and updated the model using augmented unlabeled images. Xu et al. [34] designed a Soft Teacher mechanism and a box jittering approach to select reliable pseudo boxes for training. Zhang et al. [35] mined negative samples from the images without targets and used them to update the model. In the SAR ship detection field, Wang et al. [36] designed label propagation and consistent augmentation strategies to utilize unlabeled images. Hou et al. [37] proposed semisupervised consistency learning adversarial network (SCLANet), in which an adversarial learning strategy and two pretext tasks based on consistency learning are designed for semisupervised learning. However, these methods do not take full advantage of the scene-level annotations of SAR images. How to further utilize the scene-level annotations to improve the performance of SAR ship detection is still a problem worth studying.

In addition, due to the severe clutter interference, many false alarms are often generated in SAR images, especially in inland and inshore areas. Traditional methods often perform a land–sea segmentation process [38], [39], [40] before ship detection to reduce false alarms. However, since both the ship targets and docks belong to strong scattering areas in SAR images, the land–sea segmentation methods are easy to incorporate the ship targets anchored at the dock into land areas, resulting in more missing alarms in the detection results. Moreover, the steps of land–sea segmentation methods are cumbersome and time-consuming. Therefore, designing a more precise and efficient test process is necessary to reduce the inland and inshore false alarms.

In response to the problems mentioned above, we propose our solutions in this article. To reduce the dependence of network training on expensive target-level annotations, we propose a novel semisupervised SAR ship detection network via scene characteristic learning. It can simultaneously use a large number of SAR images with only scene-level annotations and a small number of SAR images with both scene-level and target-level annotations to train. From Fig. 1, we can observe that SAR images belonging to different scenes exhibit different scene characteristics, and these differences are mainly reflected in the existence possibility of ship targets and clutter types. Combining the above scene characteristics to improve detection performance is the focus of our research. The proposed network constructs a scene characteristic learning branch parallel with the detection branch. In the scene characteristic learning branch, a scene classification loss and a scene aggregation loss are designed to utilize the scene-level annotations. These two losses can constrain the feature extraction network to learn the scene characteristics of SAR images from different perspectives. Specifically, the scene classification loss can constrain the feature extraction network to learn the distinguishable features between SAR images belonging to different scenes. The scene aggregation loss can constrain the feature extraction network to learn the consistent features between SAR images belonging to the same scene. By fully learning the scene characteristics of SAR images, the feature extraction network can enhance its feature representation ability for ship targets and clutter. Therefore, the performance of SAR ship detection can be improved. Finally, the detection loss, scene classification loss, and scene aggregation loss are jointly optimized. Compared with other semisupervised target detection methods, the proposed network makes better use of the easily obtained scene-level annotations of SAR images, thus improving the performance of SAR ship detection under limited target-level annotations.

As mentioned above, SAR images belonging to different scenes exhibit different scene characteristics. Considering that, we propose a hierarchical test process from scene to target to reduce inland and inshore false alarms. During the test process, the scene recognition process will be performed first to obtain the scene recognition results. Then, we design different detection strategies for SAR images recognized as

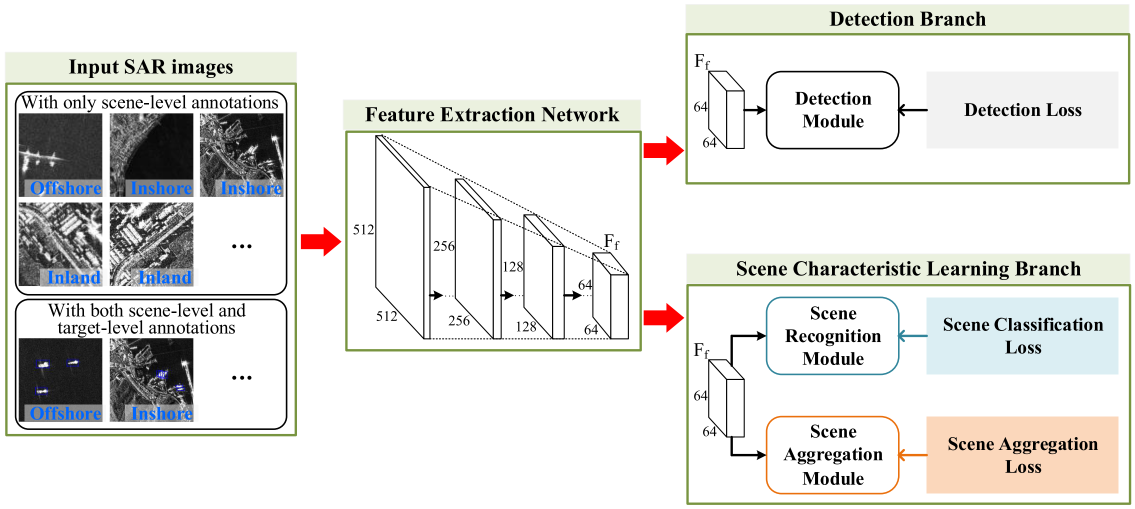


Fig. 2. Overall architecture of the proposed semisupervised SAR ship detection network. It consists of the input SAR images, the feature extraction network, the scene characteristic learning branch, and the detection branch.

the inland, inshore, and offshore scenes. More specifically, we will not perform the target detection process for the SAR images recognized as inland scenes. And for the SAR images recognized as inshore and offshore scenes, considering the different complexity of clutter in the inshore and offshore scenes, we will set two different detection thresholds to perform the target detection process. Since the scene characteristics of SAR images are fully considered, the proposed test process can significantly reduce inland and inshore false alarms while reducing the offshore missing alarms, thus improving the detection performance.

The main contributions in this article can be summarized as follows.

1) A novel semisupervised SAR ship detection network via scene characteristic learning is proposed in this article. It can improve detection performance under limited target-level annotations by utilizing scene-level annotations of SAR images. To the best of authors' knowledge, it is the first time that the scene-level annotations of SAR images have been utilized in the SAR ship detection task.
2) Designing a scene classification loss and a scene aggregation loss to utilize the scene-level annotations effectively. Under the constraint of these two losses, the feature extraction network can improve its feature representation ability for ship targets and clutter, thus improving the detection performance.
3) We propose a hierarchical test process from scene to target, in which the scene characteristics of SAR images are fully considered. The proposed test process can significantly reduce the inland and inshore false alarms while reducing offshore missing alarms.

The remainder of this article is organized as follows. Sections II and III describes the proposed method in detail. Section IV shows the experimental results and analysis based on the measured AIR-SARShip-1.0 and LS-SSDD-v1.0 datasets. And finally, Section V concludes this article.

II. Network Structure of the Proposed Method

A. Overall Framework

The overall architecture of the proposed semisupervised SAR ship detection network is illustrated in Fig. 2. As shown in Fig. 2, we adopt RefineDet [29] as our baseline. The proposed semisupervised SAR ship detection network consists of four components: the input SAR images, the feature extraction network, the detection branch, and the scene characteristic learning branch. The framework is briefly introduced as follows.

1) Input SAR Images: According to the different annotation levels, the input SAR images can be divided into two parts: the SAR images with only scene-level annotations and the SAR images with both scene-level and target-level annotations. During the training process, the SAR images of the two different annotation types are simultaneously input into the network for end-to-end training.

2) Feature Extraction Network: The feature extraction network is a truncated ResNet50 [41], which has a deep architecture to achieve a good feature representation. The feature extraction network is the basic component of the entire network, and it is applied to extract the deep features of input SAR images.

3) Detection Branch: The detection branch is added after the feature extraction network. It consists of a detection module and a detection loss. The structure of the detection module is illustrated in Fig. 3. It can extract multiscale feature maps $\boldsymbol{C}_k$ and $\boldsymbol{P}_k$, in which $k \in [1, 2, 3, 4]$ represents the kth scale feature map. Same with RefineDet, the multiscale feature fusion operation and the two-step cascaded classification and regression process are both performed in the detection module. The multiscale fusion operation can integrate the large-scale context from $\boldsymbol{C}_k$, and convert it to $\boldsymbol{P}_k$. It is beneficial for detecting ship targets of different sizes. The two-step cascaded classification and regression process can mitigate the imbalance of positive and negative examples and improve the locating accuracy of ship targets. Specifically, the preset

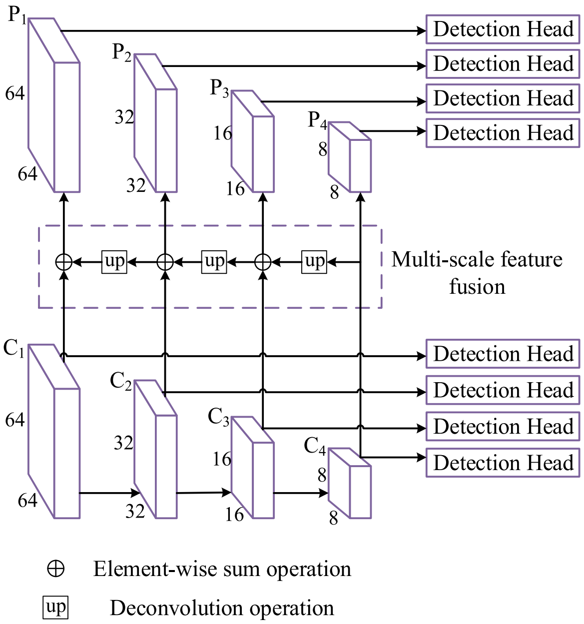


Fig. 3. Structure of the detection module.

anchor boxes and feature map $\boldsymbol{C}_k$ are first fed into the detection head to perform the first-step classification and regression. This process can obtain the rough bounding boxes. Then, the rough bounding boxes with high confidence scores and feature map $\boldsymbol{P}_k$ are fed into the detection head to perform the second-step classification and regression. This process can obtain the predicted fine bounding boxes. More details about the detection module can also be traced in [29].

The detection loss is defined as follows:

$$L_{\mathrm{DET}} = \frac{1}{N_1}\left[\sum_b L_{\mathrm{cls}}\left(p_{b1}, p_b^*\right) + \sum_b p_b^* L_{\mathrm{reg}}\left(t_{b1}, t_b^*\right)\right] + \frac{1}{N_2}\left[\sum_b L_{\mathrm{cls}}\left(p_{b2}, p_b^*\right) + \sum_b p_b^* L_{\mathrm{reg}}\left(t_{b2}, t_b^*\right)\right] \quad (1)$$

where N_1 denotes the number of anchor boxes in the first step. p_{b1} and p_b^* represent the predicted probability and the ground truth, respectively, of the bth anchor box. t_{b1} is the regression target of the bth anchor box, and t_b^* represents the true bounding box coordinates. N_2, p_{b2}, and t_{b2} are the symbols in the second step, and their definitions are similar to those of N_1, p_{b1}, and t_{b1}. The classification loss L_{cls} is the cross-entropy, and the regression loss L_{reg} is the smooth L1 loss.

4) Scene Characteristic Learning Branch: The scene characteristic learning branch, which is parallel with the detection branch, is also added after the feature extraction network. The scene characteristic learning branch is one of the core components of the proposed network. It aims to utilize the scene-level annotations of SAR images to improve the detection performance with limited target-level annotations.

Among the four components mentioned above, the details of the proposed scene characteristic learning branch are introduced concretely in Section II-B.

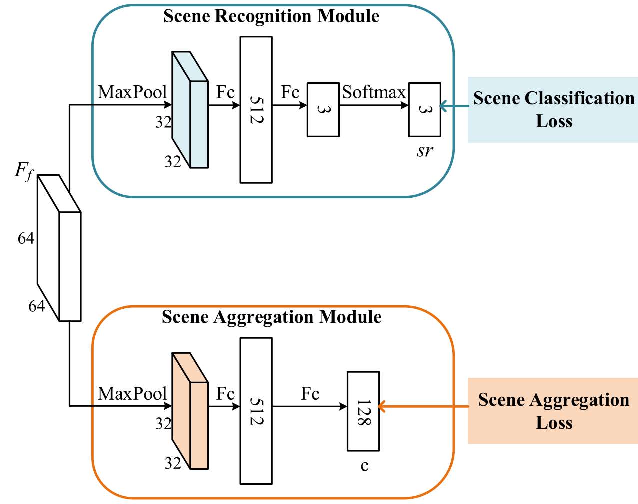


Fig. 4. Structure of the scene characteristic learning branch.

B. Scene Characteristic Learning Branch

In the SAR ship detection task, the scenes of SAR images can be mainly classified into the following three categories: inland scene, inshore scene, and offshore scene. SAR images belonging to the three scenes exhibit different scene characteristics in terms of the existence possibility of ship targets and clutter types. More specifically, in terms of the existence possibility of ship targets, the ship targets may appear in the inshore and offshore scenes but never appear in the inland scenes. In terms of the clutter types, the clutter in the inland and inshore scenes is mainly composed of the buildings, containers, harbor facilities, and other man-made complex clutter. The clutter in the offshore scene is mainly composed of the reef, sea clutter, and other relatively simple natural clutter. It makes us naturally think that the process of learning the above scene characteristics is the process of learning good feature representations for ship targets and clutter, which is beneficial to SAR ship detection. Based on this inspiration, we designed our scene characteristic learning branch. The structure of the scene characteristic learning branch is illustrated in Fig. 4. It comprises the scene recognition module, the scene aggregation module, the scene classification loss, and the scene aggregation loss. The scene classification loss and scene aggregation loss can constrain the feature extraction network to learn the scene characteristics of SAR images from different perspectives. The details of these four components are introduced as follows.

1) Scene Recognition Module and Scene Classification Loss: The scene recognition module is applied to recognize the scene type of the input SAR image. The scene recognition module takes the deep feature F_{f} obtained by the feature extraction network as input, and it outputs the scene recognition result of the input SAR image.

In the scene recognition module, the max-pooling operation is first adopted along the spatial dimension to downsample the input feature F_{f}. Then, two fully connected layers and one softmax classifier are adopted to produce the scene recognition result sr. The computation process can be summarized as follows:

$$\mathrm{sr} = \mathrm{Softmax}(Fc_r(\mathrm{MaxPool}(F_\mathrm{f}))) \tag{2}$$

where MaxPool represents the max-pooling, Fc_r represents the two fully connected layers in the scene recognition module, and Softmax represents the softmax function.

The scene classification loss is the cross-entropy loss function. It is calculated by the scene recognition result sr and the scene-level annotations. The definition is as follows:

$$L_{\mathrm{SC}} = -\frac{1}{N}\sum_{i=1}^{N}\left[\mathrm{sr}_i \log\left(\mathrm{sr}_i^*\right) + (1-\mathrm{sr}_i)\log\left(1-\mathrm{sr}_i^*\right)\right] \tag{3}$$

where N denotes the number of input SAR images in a mini-batch. sr_i and sr_i^* denote the scene recognition result and the scene-level annotation, respectively, of the ith input SAR image.

Under the constraint of scene classification loss, the scene recognition module can have the ability to identify the scene types of SAR images, and the feature extraction network can learn the distinguishable features between SAR images belonging to different scenes. In this way, the scene characteristics of SAR images can be learned.

2) Scene Aggregation Module and Scene Aggregation Loss: The scene aggregation module also takes the deep feature F_f obtained by the feature extraction network as input. First, the max-pooling operation is adopted along the spatial dimension to downsample the input feature F_f. Then, two fully connected layers are used to project the F_f to a 128-D embedding space. Thus, the embedded feature z, which can be regarded as the feature representation of the input SAR image in the embedding space, can be obtained. The computation process of the scene aggregation module is summarized as follows:

$$z = Fc_a(\mathrm{MaxPool}(F_\mathrm{f})) \tag{4}$$

where Fc_a denotes the two fully connected layers in the scene aggregation module.

Chen et al. [30] and He et al. [31] learn good representations by maximizing agreement between two positive pairs. Inspired by these methods, we designed our scene aggregation loss. The core idea of this loss is to maximize the feature similarity between the SAR images belonging to the same scene in the embedding space. The definition is as follows:

$$L_{\mathrm{SA}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{m=1}^{N}\omega_{mi} I_{[m\neq i]} I_{[\mathrm{sr}_m^*=\mathrm{sr}_i^*]} \times \log\frac{\exp\left(\frac{s(z_i,z_m)}{\tau}\right)}{\sum_{n=1}^{N} I_{[n\neq i]}\exp\left(\frac{s(z_i,z_n)}{\tau}\right)} \tag{5}$$

where $I_{[m\neq i]} \in \{0,1\}$, $I_{[\mathrm{sr}_m^*=\mathrm{sr}_i^*]} \in \{0,1\}$, and $I_{[n\neq i]} \in \{0,1\}$ are the indicator functions

$$I_{[m\neq i]} = \begin{cases}1, & \text{if } m \neq i\\ 0, & \text{otherwise}\end{cases},\quad I_{[\mathrm{sr}_m^*=\mathrm{sr}_i^*]} = \begin{cases}1, & \text{if } \mathrm{sr}_m^* = \mathrm{sr}_i^*\\ 0, & \text{otherwise}\end{cases} \tag{6}$$

$$I_{[n\neq i]} = \begin{cases}1, & \text{if } n \neq i\\ 0, & \text{otherwise.}\end{cases} \tag{7}$$

The $s(u,v)$ in (5) denotes the dot product between l_2 normalized u and v (i.e., cosine similarity)

$$s(u,v) = \frac{u^{\mathrm{T}}v}{\|u\|\|v\|}. \tag{8}$$

The τ in (5) denotes a scalar temperature parameter. The ω_{mi} is a scalar weight that needs to be set manually. It controls the degree of pulling close the embedded feature z_i and z_m. Please note that the input SAR images contain the augmented SAR images. And the augmented SAR image is identical to its original SAR image in terms of ship targets and clutter types. Therefore, the augmented SAR images from the same original SAR image should be closer to each other than other SAR images in the embedded space. Based on this idea, when the mth and ith input SAR images are from the same original SAR image, a relatively large weight is set, i.e., $\omega_{mi} = 0.8$; otherwise, a relatively small weight is set, i.e., $\omega_{mi} = 0.2$.

Under the constraint of scene aggregation loss, SAR images belonging to the same scene can be pulled close to each other in the embedded space. In this way, the feature extraction network can learn the consistent features between SAR images belonging to the same scene. Thus, the scene characteristics of SAR images can be further learned.

III. Algorithm Flow of the Proposed Method

In this section, the semisupervised training process and the hierarchical test process from scene to target are introduced concretely.

A. Semisupervised Training Process

Unlike supervised learning, the semisupervised training process of the proposed network utilizes a large number of SAR images with only scene-level annotations and a small number of SAR images with both scene-level and target-level annotations. The entire network is jointly trained with SAR images of the two different annotation types during the training. The total loss of the network consists of the scene classification loss, the scene aggregation loss, and the detection loss. The scene classification loss and scene aggregation loss need to use the scene-level annotations of SAR images. And the detection loss needs to use the target-level annotations of SAR images. The total loss is defined as follows:

$$L = \lambda_1 L_{\mathrm{SC}} + \lambda_2 L_{\mathrm{SA}} + L_{\mathrm{DET}} \tag{9}$$

where λ_1 and λ_2 are the weight factors that balance the L_{SC}, L_{SA}, and L_{DET}.

The entire network is jointly optimized in an end-to-end manner. Through this semisupervised training method, the dependence of network training on the expensive target-level annotations of SAR images can be reduced.

B. Hierarchical Test Process From Scene to Target

Fig. 5 and Algorithm 1 show the flowchart and pseudocode of the proposed hierarchical test process from scene to target. After the training of the entire network is completed, the proposed test process calls the trained network to detect the

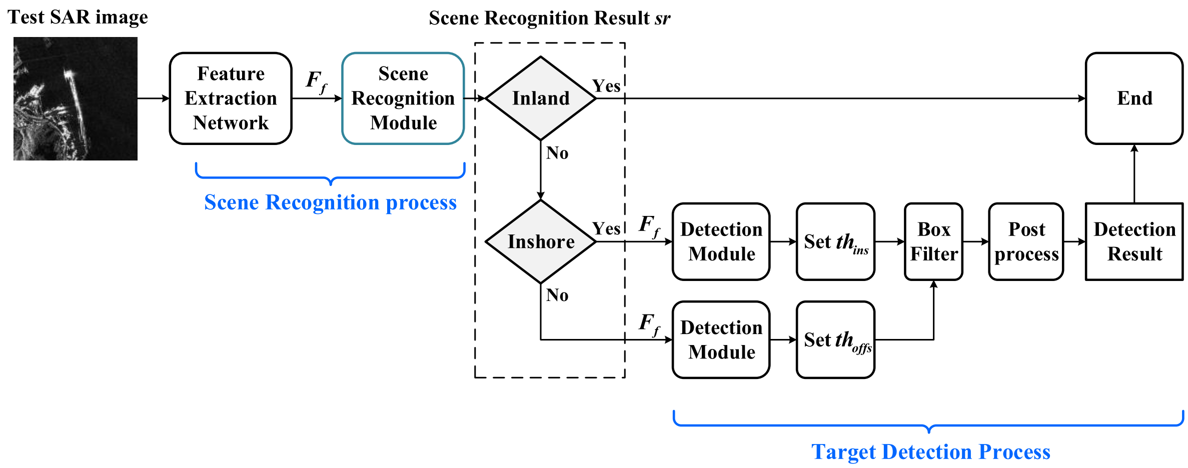


Fig. 5. Flowchart of the proposed hierarchical test process from scene to target.

Algorithm 1 The Hierarchical Test Process from Scene to Target

Input: test SAR image x, the trained network $net(\cdot)$, the detection threshold for the inshore scene th_{ins}, the detection threshold for the offshore scene th_{offs};
Output:the detection result dr;

$F_f \leftarrow net_{FEM}(x)$;
$sr \leftarrow net_{SRM}(F_f)$;
if $sr ==$ '$inland'$ **then**
  $dr = \emptyset$;
else if $sr ==$ '$inshore'$ **then**
  $b \leftarrow net_{DM}(F_f)$;
  $dr \leftarrow NMS(BoxFilter(b, th_{ins}))$;
else then
  $b \leftarrow net_{DM}(F_f)$
  $dr \leftarrow NMS(BoxFilter(b, th_{offs}))$
end if
return dr

test SAR image. And please note that the trained network only needs to be called once during the whole test process. The proposed test process can be roughly divided into the scene recognition process and the target detection process. The specific process is as follows.

1) Scene Recognition Process: The SAR image will be first input into the feature extraction network to obtain the deep feature F_f. Then, the F_f will be input into the scene recognition module to obtain the scene recognition result sr.

2) Target Detection Process: Considering the prior information that the ship targets never appear in the inland areas, we only perform the target detection process for the SAR images recognized as inshore and offshore scenes. In the target detection process, the deep feature F_f of the test SAR image will be first input into the detection module to obtain the predicted bounding boxes. If the test SAR image is recognized as the inshore scene, we will set the threshold th_{ins} to filter out the bounding boxes. If the test SAR image is recognized as the offshore scene, we will set the threshold th_{offs} to filter out the bounding boxes. The traditional approach generally sets a single detection threshold to filter out the bounding boxes that are less likely to be true targets. However, due to the different complexity of clutter, the optimal detection thresholds for the inshore and offshore scenes should not be equal. Intuitively, the inshore scene contains more complex man-made clutter than the offshore scene. Thus, a relatively higher detection threshold can be set for the inshore scene to ensure fewer false alarms, and a relatively lower detection threshold can be set for the offshore scene to ensure fewer missing alarms. We determine the detection thresholds for the inshore scene (i.e., the th_{ins}) and offshore scene (i.e., the th_{offs}) through a cross-validation process. Finally, after the bounding boxes are filtered, we perform the postprocess, i.e., nonmaximum suppression (NMS) [42], to obtain the final detection results.

The traditional test process does not recognize the scene types of the test SAR images and adopts the same detection strategy for SAR images belonging to different scenes. Compared with that, the hierarchical test process from scene to target fully considers the scene characteristic of SAR images, making the design of detection strategies more targeted and flexible. The proposed test process can significantly reduce inland and inshore false alarms while reducing offshore missing alarms. It is conducive to improve the overall detection performance.

IV. Experimental Results and Analysis

A. Datasets

1) AIR-SARShip-1.0: The AIR-SARShip-1.0 dataset was released by Sun et al. [43], and it has been widely used for SAR ship detection [18], [19], [20]. The AIR-SARShip-1.0 dataset is derived from the Gaofen-3 satellite and contains 31 large-scale SAR images. The resolutions of these SAR images vary from 1 to 3 m, and the sizes are 3000 × 3000 pixels. The imaging modes include spotlight and stripe. The polarization mode is single polarization. Among the 31 SAR images, 21 images are used for training and validation, and the remaining ten images are used for testing. In our experiment, the sizes of the original large-scale SAR images

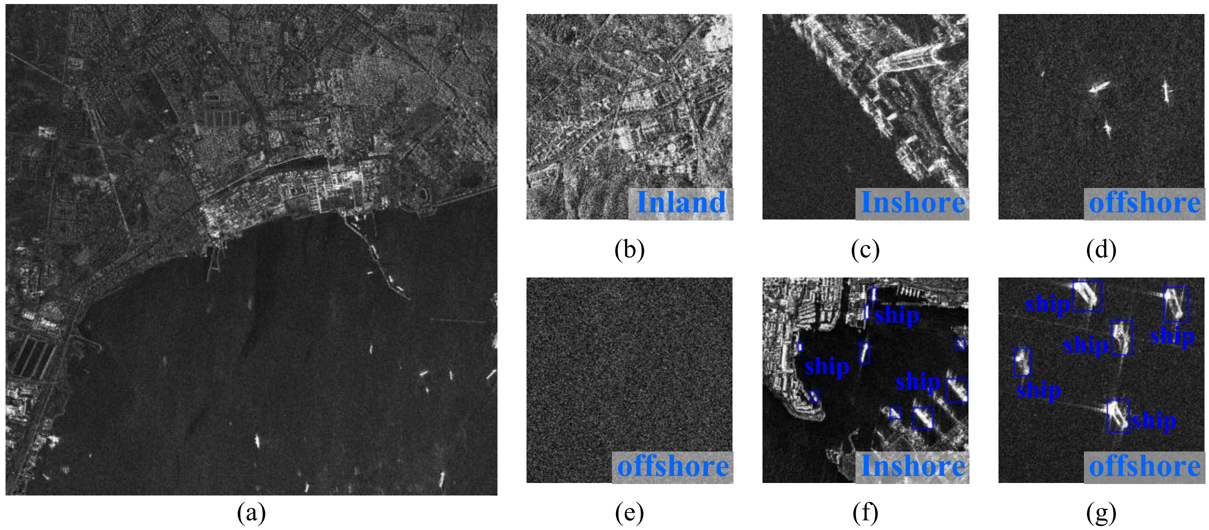


Fig. 6. Several SAR images from the AIR-SARShip-1.0 dataset: (a) example of the large-scale SAR image; (b)–(e) some SAR subimages with only scene-level annotations; and (f) and (g) some SAR subimages with both scene-level and target-level annotations.

are too large, not satisfying the size restriction of general network inputs. Therefore, a series of 512 × 512 subimages is cropped from the original large-scale SAR images by sliding window with an overlap of 100. And the total subimage number is 1984.

2) LS-SSDD-v1.0: To further confirm the effectiveness of the proposed method, we also conduct experiments on the LS-SSDD-v1.0 dataset. The LS-SSDD-v1.0 dataset was released by Zhang et al. [44], and it has already been used by many scholars [9], [10], [15]. The LS-SSDD-v1.0 is composed of 15 large-scale Sentinel-1 images in the interferometric wide-swath (IW) mode. The resolutions of these SAR images are 5 × 20 m, and the sizes are 24 000 × 16 000 pixels. Among the 15 SAR images, ten images are used for training and validation, and the remaining five images are used for testing. In our experiment, we cropped the original large-scale SAR images into 800 × 800 subimages by the sliding window without overlapping, keeping to the method of Zhang et al. [44]. And the total subimage number is 9000.

For the AIR-SARShip-1.0 and LS-SSDD-v1.0 datasets, please note that all training subimages are assigned the scene-level annotations, whether they contain the ship targets or not. And to validate the detection performance of the proposed method in the case of limited target-level annotations, we select 30% of training subimages that contain ship targets to assign the target-level annotations further, and the remaining 70% still only have the scene-level annotations. After the 30% of training subimages with target-level annotations are selected, we augment all training subimages by adding noise and rotating. Please note that the 30% of training subimages with target-level annotations are randomly selected five times, and the average result of the five random experiments is taken as the final result. During the test process, all the test subimages are fed into the network to process individually. After that, the prediction results of all test subimages are restored to the original SAR image. Finally, the NMS algorithm with the intersection over union (IoU) threshold of 0.2 is employed to remove the overlapped bounding boxes.

Fig. 6 shows a large-scale SAR image and several subimages of the AIR-SARShip-1.0. Fig. 7 shows a large-scale SAR image and several subimages of the LS-SSDD-v1.0.

B. Experimental Settings

The experimental hyperparameters are set as follows: the weight factors λ_1 and λ_2 in (9) are both set to 0.1. We use the stochastic gradient descent (SGD) optimizer as the training optimizer, the weight decay is 0.0005, and the momentum is 0.9. The mini-batch size is 12, and the initial learning rate is 0.001. The network is trained by 120k iterations in total, and the learning rate is divided by a factor of 10 at the 80k and 100k iterations. After cross-validation, the detection thresholds $\mathrm{th}_{\mathrm{ins}}$ and $\mathrm{th}_{\mathrm{offs}}$ are set to 0.9 and 0.7, respectively, for the AIR-SARShip-1.0 dataset. For the LS-SSDD-v1.0 dataset, the detection thresholds $\mathrm{th}_{\mathrm{ins}}$ and $\mathrm{th}_{\mathrm{offs}}$ are set to 0.5 and 0.4, respectively. The experiments are implemented with the deep learning framework Pytorch [45], using a personal computer with Intel Xeon E5-2630 v4 CPU and NVIDIA GeForce GTX 1080Ti GPU on a Ubuntu 16.04 Linux system.

C. Evaluation Criteria

We quantitatively evaluate the detection performance of different detection methods via three widely used metrics, namely, precision, recall, and F1-score. Their definitions are as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{10}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{11}$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision}+\text{Recall}} \tag{12}$$

where TP (true positives) is the number of correctly detected ship targets, FP (false positives) is the number of false alarms,

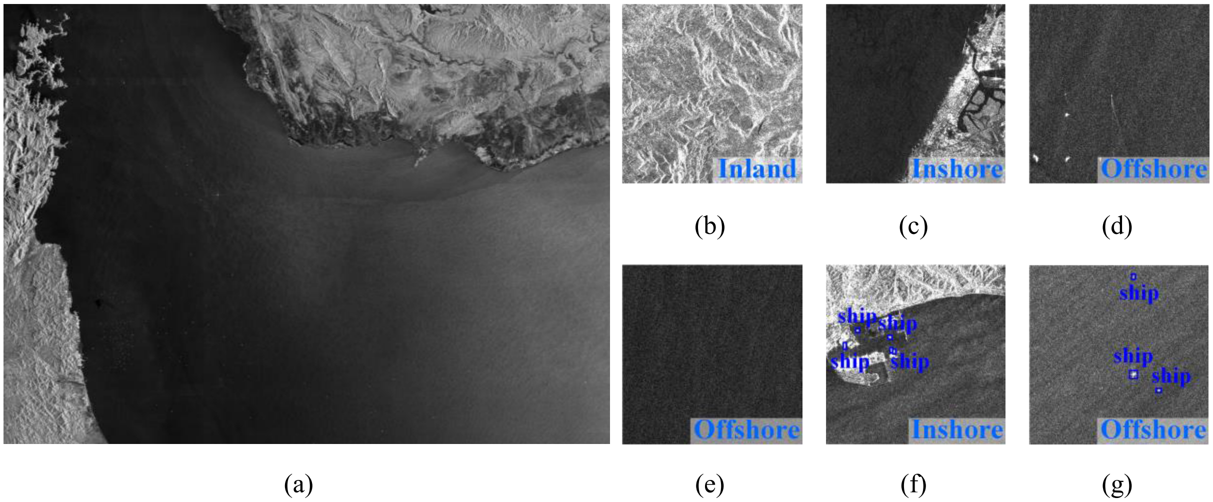


Fig. 7. Several SAR images from the LS-SSDD-v1.0 dataset: (a) example of the large-scale SAR image; (b)–(e) some SAR subimages with only scene-level annotations; and (f) and (g) some SAR subimages with both scene-level and target-level annotations.

TABLE I
QUANTITATIVE EXPERIMENTAL RESULTS OF DIFFERENT METHODS UNDER THE CONDITION THAT THE TRAINING SAMPLES WITH TARGET-LEVEL ANNOTATIONS ACCOUNT FOR 30% OF ALL TRAINING SAMPLES CONTAINING SHIP TARGETS

Method	Supervision mode	AIR-SARShip-1.0 dataset			LS-SSDD-v1.0 dataset		
		Precision (%)	Recall (%)	F1-score (%)	Precision (%)	Recall (%)	F1-score (%)
SSD	Fully supervised	75.42	72.00	73.27	43.05	52.57	47.33
RetinaNet		78.39	72.80	75.49	50.75	58.87	54.51
Faster R-CNN		73.43	84.80	78.70	63.56	59.71	61.58
RefineDet		72.97	87.20	79.45	66.72	70.23	68.43
YOLOX		81.51	78.40	79.93	66.78	**75.44**	70.85
STAC	Semi-supervised	76.06	85.60	80.55	73.12	61.02	66.52
Soft Teacher		78.46	84.00	81.14	71.11	66.65	68.81
Proposed method		**83.72**	**88.80**	**86.18**	**85.57**	73.72	**79.20**

and FN (false negatives) is the number of missing alarms. The precision measures the fraction of TPs among all detected results. The recall measures the fraction of TPs over the number of ground truths. The F1-score is the harmonic mean between the precision and recall, and it can comprehensively evaluate the quality of a detection method. The higher the value of the above three evaluation criteria, the better the performance of the detection method is.

D. Comparison With Other Detection Methods

In this section, we compare the proposed semisupervised SAR ship detection method with other famous deep-learning-based detection methods, including some fully supervised detection methods and some semisupervised detection methods. The fully supervised detection methods for comparison include three classic detection methods, i.e., the SSD [28], RetinaNet [46], and Faster R-CNN [27], and other two detection methods which have been proposed in the last four years, i.e., the RefineDet [29] and YOLOX [47]. The semisupervised detection methods for comparison include two methods based on self-training, i.e., the STAC [33] and soft-teacher [34], which have been proposed in the last two years. In the case that the training samples with target-level annotations account for 30% of all training samples containing ship targets, Fig. 8 shows the detection results of different methods on three large-scale test SAR images of AIR-SARShip-1.0, Fig. 9 shows the detection results of different methods on a large-scale test SAR image of LS-SSDD-v1.0, and Table I shows the quantitative experimental results on the two datasets.

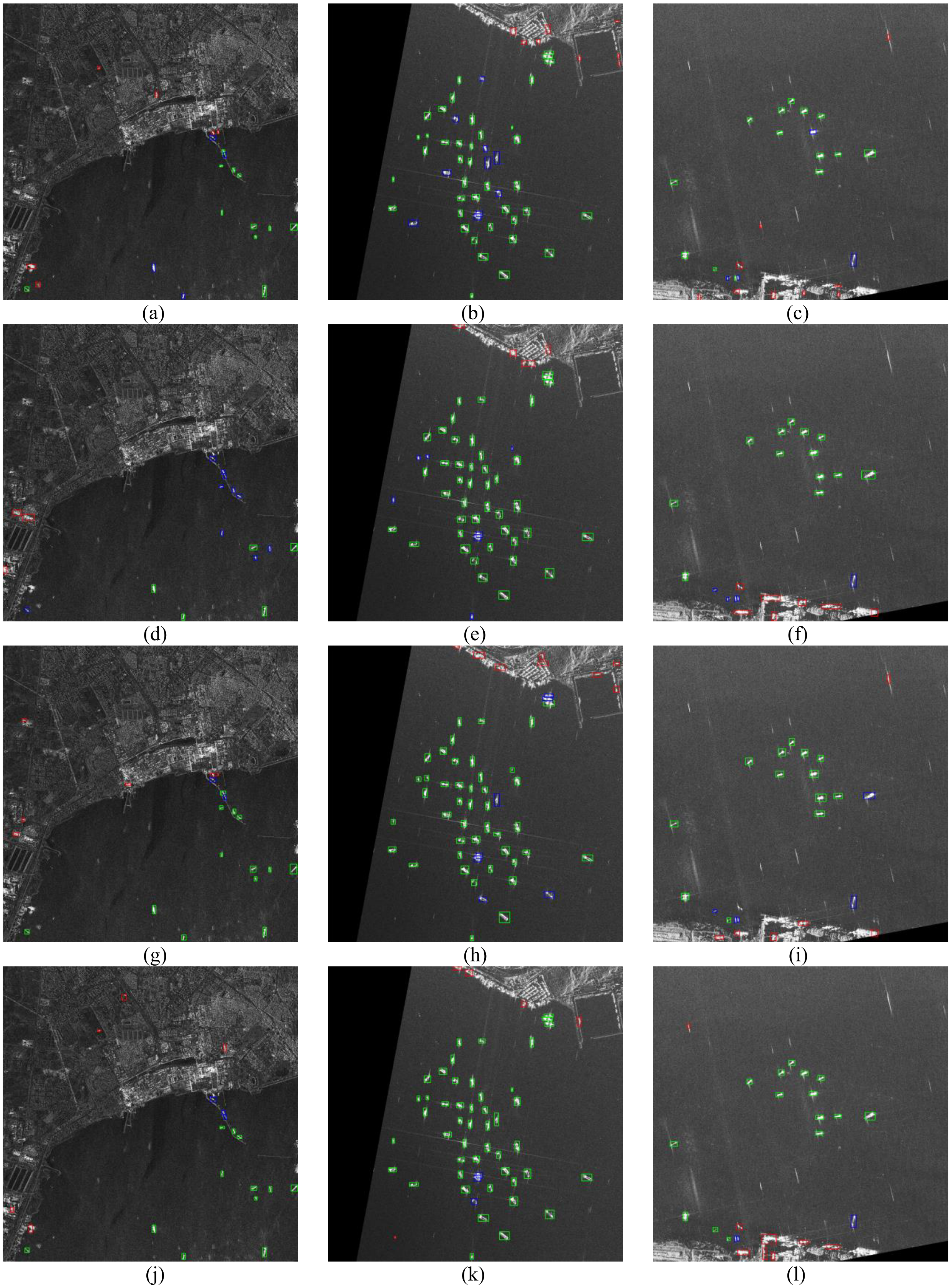


Fig. 8. Detection results compared with other detection methods on three large-scale test SAR images of AIR-SARShip-1.0, where green rectangles represent the correctly detected target chips, red rectangles represent false alarms, and blue rectangles represent the missing alarms: (a)–(c) SSD; (d)–(f) RetinaNet; (g)–(i) Faster R-CNN; (j)–(l) RefineDet; (m)–(o) YOLOX; (p)–(r) STAC; (s)–(u) Soft Teacher; and (v)–(x) proposed method.

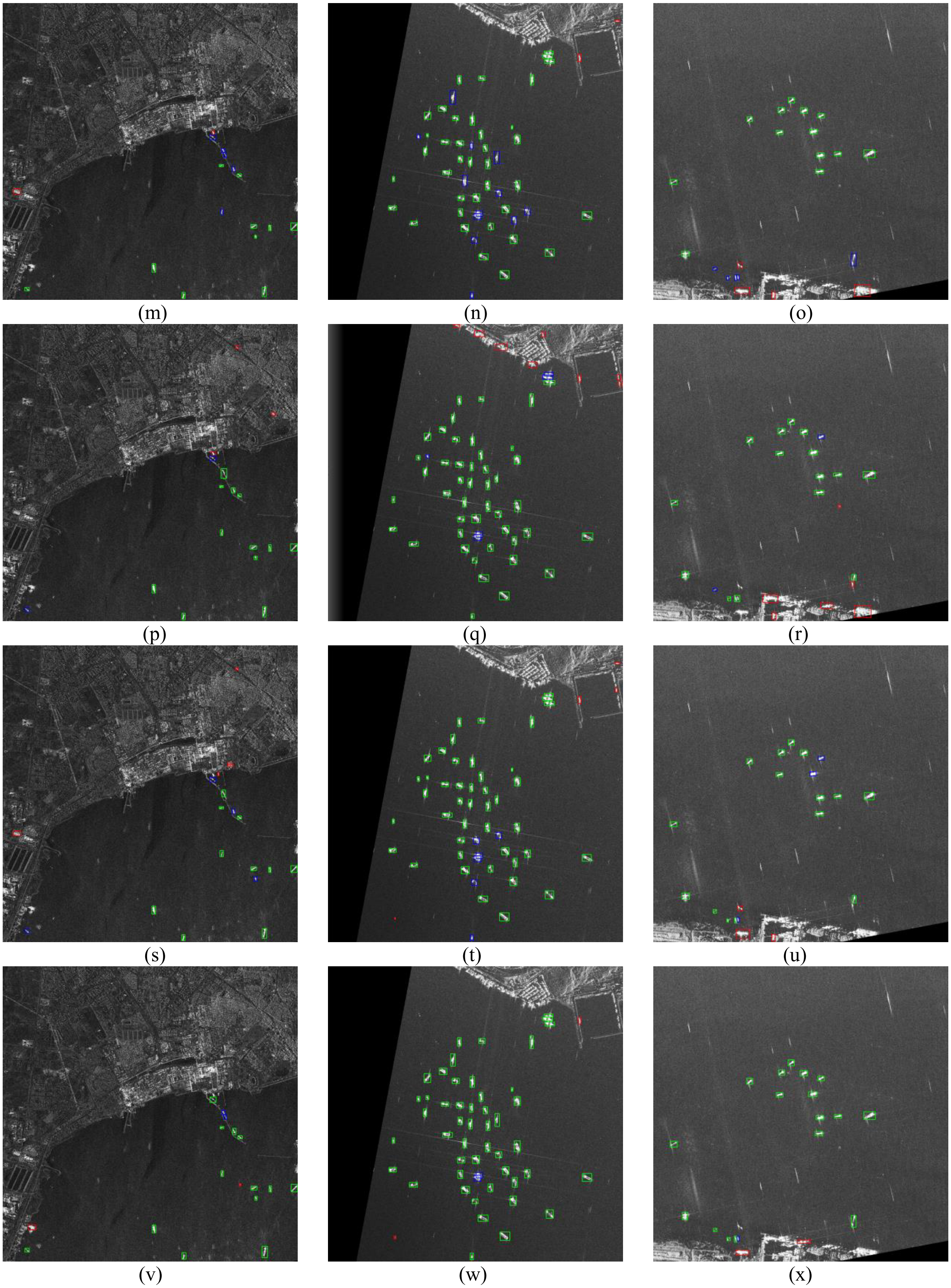
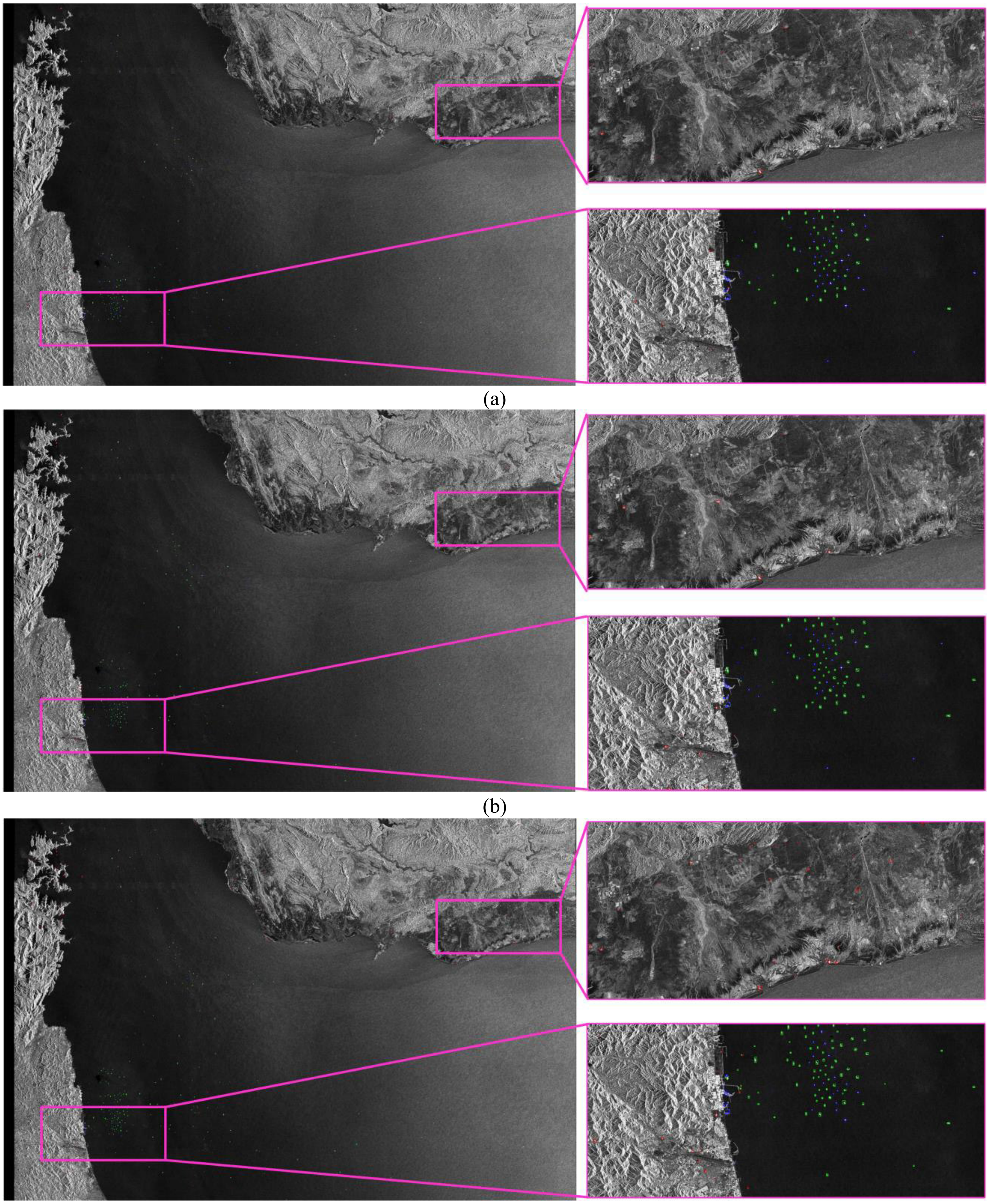


Fig. 8. *(Continued.)* Detection results compared with other detection methods on three large-scale test SAR images of AIR-SARShip-1.0, where green rectangles represent the correctly detected target chips, red rectangles represent false alarms, and blue rectangles represent the missing alarms: (a)–(c) SSD; (d)–(f) RetinaNet; (g)–(i) Faster R-CNN; (j)–(l) RefineDet; (m)–(o) YOLOX; (p)–(r) STAC; (s)–(u) Soft Teacher; and (v)–(x) proposed method.

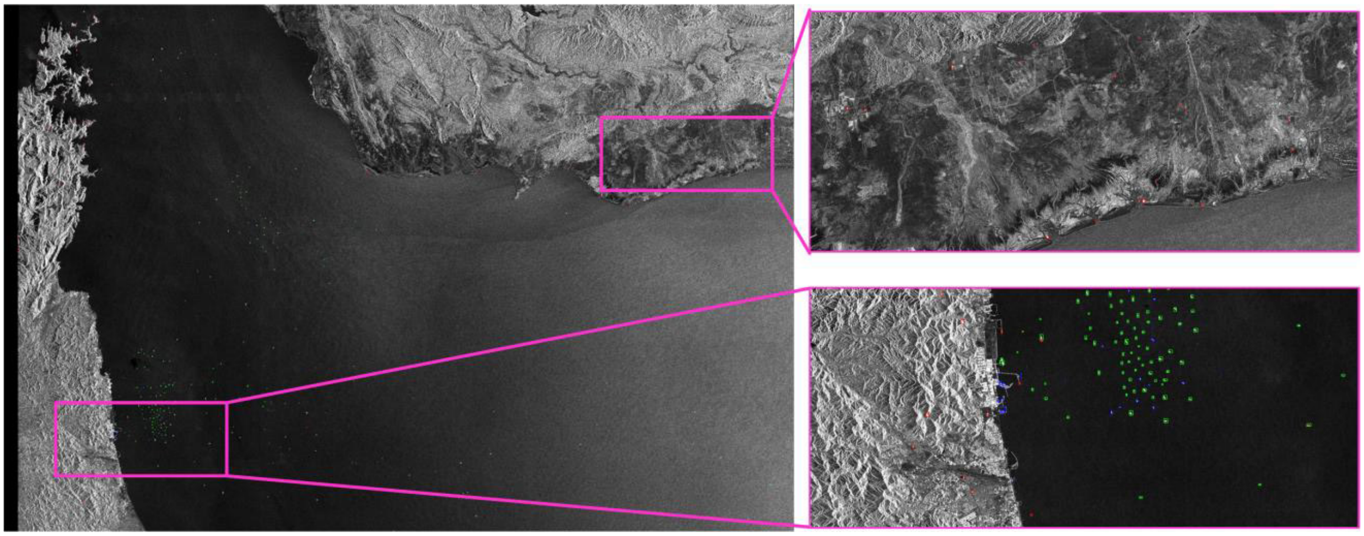


(a)

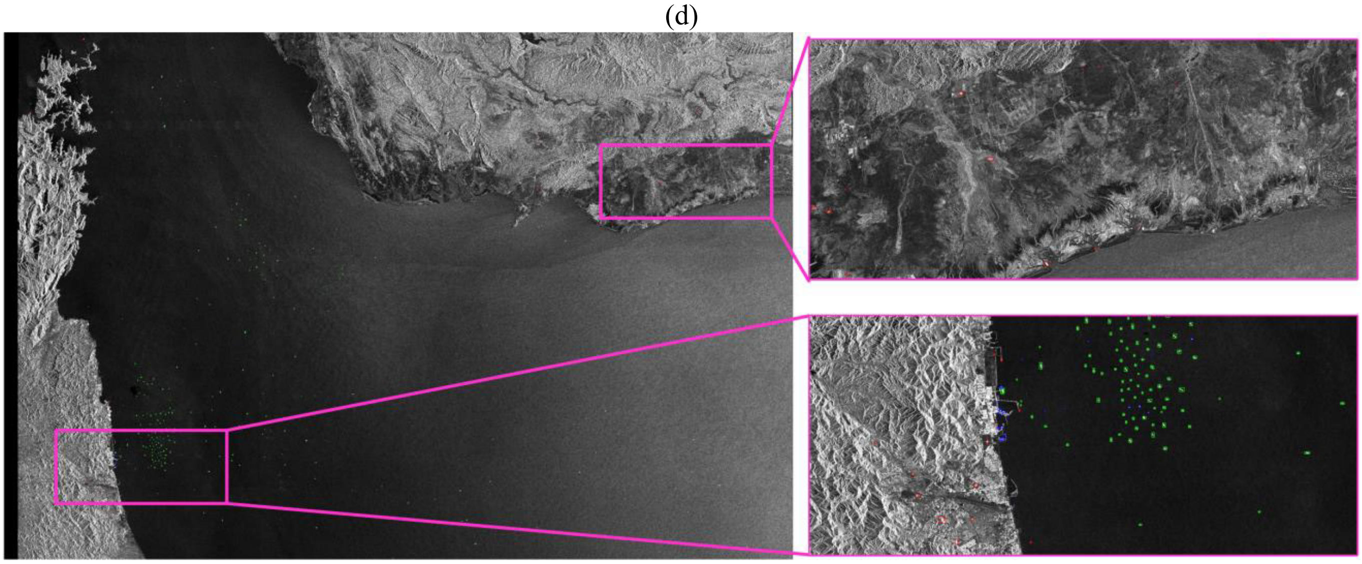
(b)

(c)

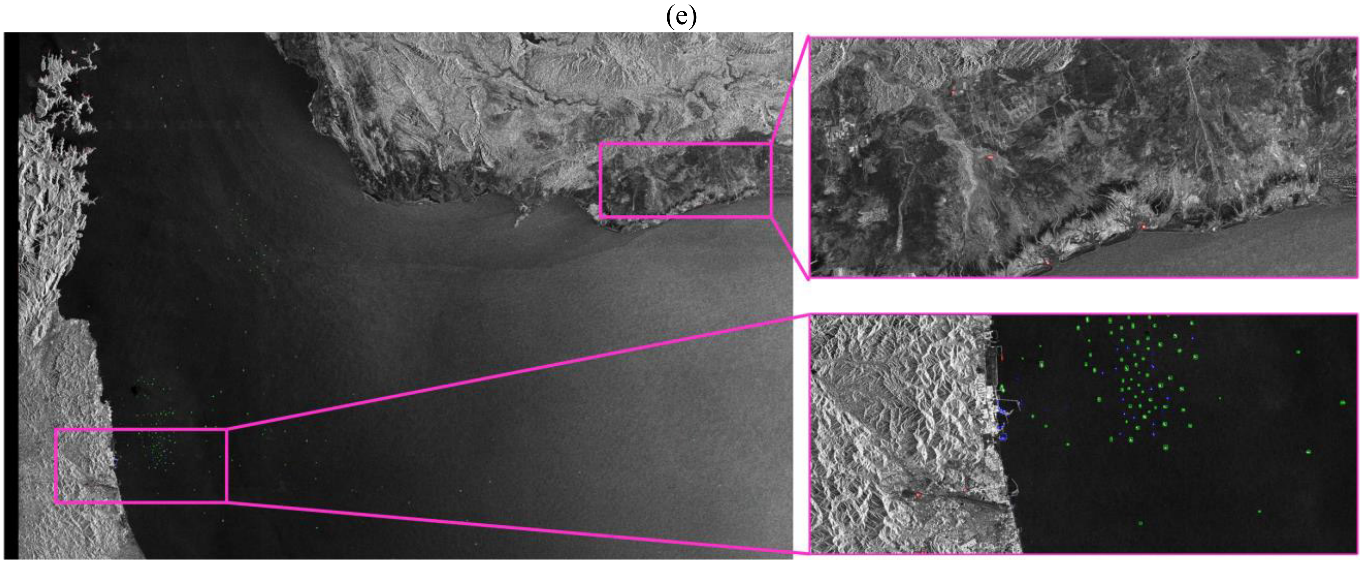
Fig. 9. Detection results compared with other detection methods on a large-scale test SAR image of LS-SSDD-v1.0, where green rectangles represent the correctly detected target chips, red rectangles represent false alarms, and blue rectangles represent the missing alarms: (a) SSD; (b) RetinaNet; (c) Faster R-CNN; (d) RefineDet; (e) YOLOX; (f) STAC; (g) Soft Teacher; and (h) proposed method.



(d)

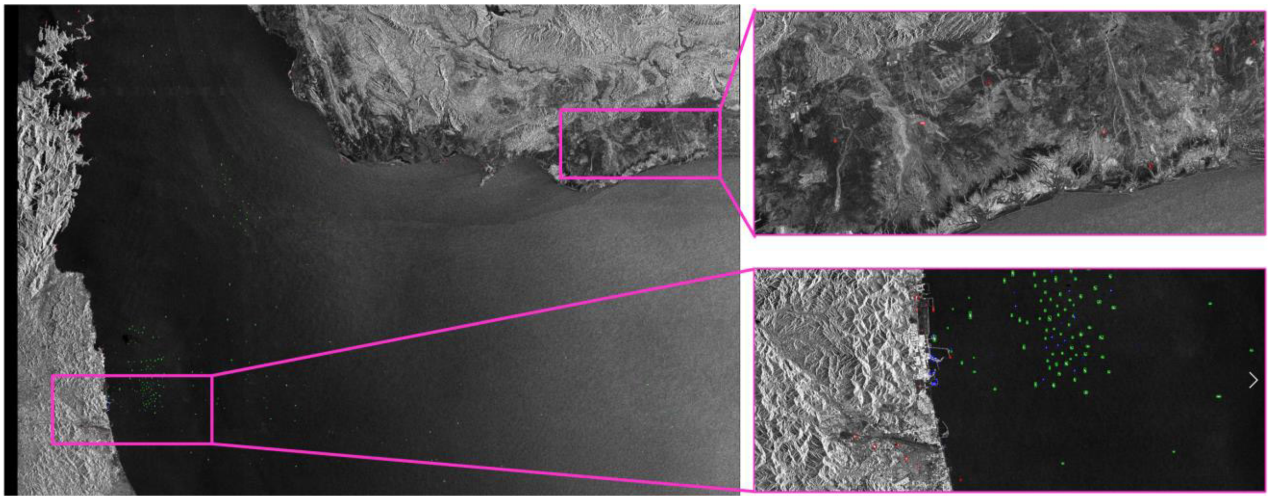


(e)

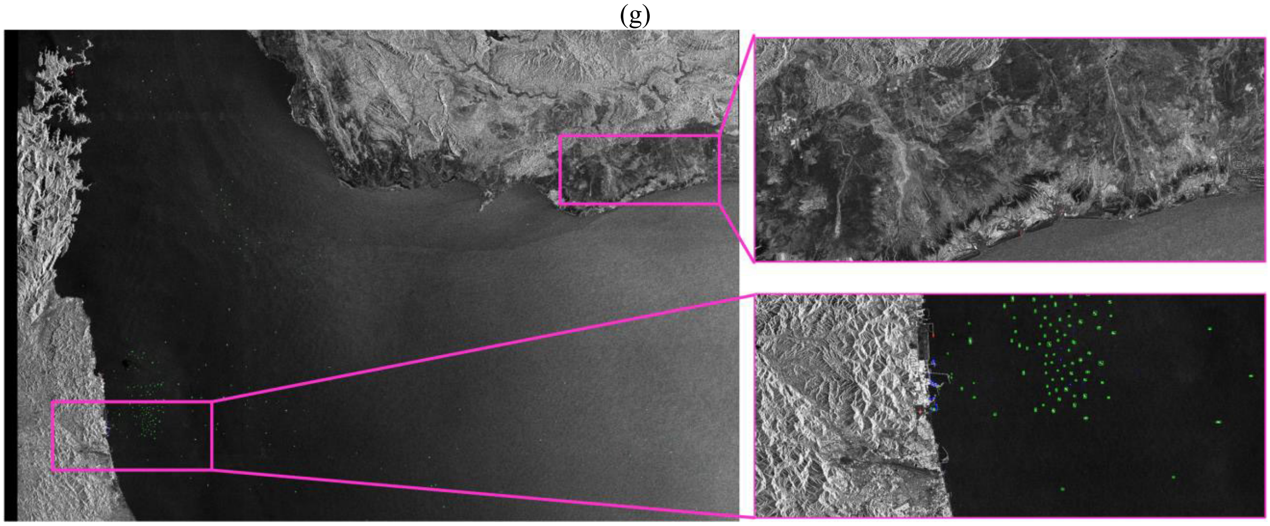


(f)

Fig. 9. *(Continued.)* Detection results compared with other detection methods on a large-scale test SAR image of LS-SSDD-v1.0, where green rectangles represent the correctly detected target chips, red rectangles represent false alarms, and blue rectangles represent the missing alarms: (a) SSD; (b) RetinaNet; (c) Faster R-CNN; (d) RefineDet; (e) YOLOX; (f) STAC; (g) Soft Teacher; and (h) proposed method.



(g)



(h)

Fig. 9. *(Continued.)* Detection results compared with other detection methods on a large-scale test SAR image of LS-SSDD-v1.0, where green rectangles represent the correctly detected target chips, red rectangles represent false alarms, and blue rectangles represent the missing alarms: (a) SSD; (b) RetinaNet; (c) Faster R-CNN; (d) RefineDet; (e) YOLOX; (f) STAC; (g) Soft Teacher; and (h) proposed method.

As shown in Figs. 8 and 9, we can see that the detection results of SSD and RetinaNet, i.e., Figs. 8(a)–(f) and 9(a) and (b), show a lot of missing alarms in the offshore scene and a lot of false alarms in the inland and inshore scenes. Compared with SSD and RetinaNet, the Faster R-CNN, RefineDet, and YOLOX have better detection performance, but they also produce many false alarms and missing alarms, as shown in Figs. 8(g)–(o) and 9(c)–(e). In the semisupervised detection methods, both STAC and Soft Teacher are based on Faster R-CNN. Due to the use of unlabeled training samples, these two methods achieved relatively better detection results than Faster R-CNN, as shown in Figs. 8(p)–(u) and 9(f) and (g), respectively. However, a large number of false alarms are still generated in the strong scattering area of inland and inshore scenes. Obviously, the proposed method achieved the best detection results compared with other detection methods, as shown in Figs. 8(v)–(x) and 9(h). The proposed method produces no false alarms in the inland areas, a small number of false alarms in the inshore areas, and a small number of missing alarms in the offshore areas.

As shown in Table I, we list the quantitative experimental results of different detection methods on two datasets. Please note that the training samples with target-level annotations account for only 30% of all training samples containing ship targets. Since the fully supervised methods can only use these training samples with target-level annotations to train, their detection performance is relatively poor. Nevertheless, compared with most fully supervised methods, the STAC and Soft Teacher can achieve higher precision, recall, and F1-score, but the performance improvement is limited. For the proposed method, the training samples with only scene-level annotations can also be used to train the network. And benefits from the effective use of scene-level annotations, we can find that the proposed method can achieve the highest F1-score, which are 86.18% on the AIR-SARShip-1.0 and 79.2% on the LS-SSDD-v1.0. Therefore, the experimental results in Table I proves that the proposed method can improve the detection performance in the case of limited target-level annotations by utilizing scene-level annotations.

TABLE II
ABLATION EXPERIMENTS UNDER THE CONDITION THAT THE TRAINING SAMPLES WITH TARGET-LEVEL ANNOTATIONS ACCOUNT FOR 30% OF ALL TRAINING SAMPLES CONTAINING SHIP TARGETS

Components?			Precision (%)	Recall (%)	F1-score (%)
Scene Characteristic Learning Branch		Hierarchical Test Process from Scene to Target			
Scene Classification Loss	Scene Aggregation Loss				
×	×	×	72.97	87.20	79.45
×	✓	×	79.11	87.20	82.95
✓	×	×	76.39	90.40	82.81
✓	✓	×	81.06	88.00	84.38
✓	×	✓	78.87	**91.20**	84.59
✓	✓	✓	**83.72**	88.80	**86.18**

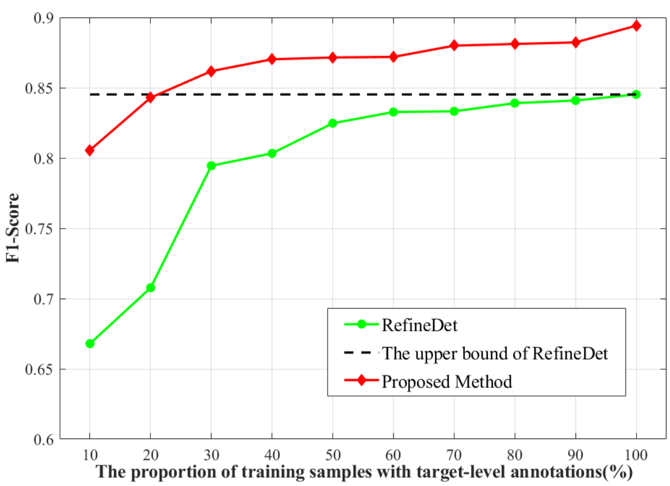


Fig. 10. Detection performance of the proposed method under different proportions of training samples with target-level annotations.

E. Model Analysis

1) Analysis of the Detection Performance Under Different Proportions of Training Samples With Target-Level Annotations: To analyze the detection performance of the proposed method under different proportions of training samples with target-level annotations, in this section, we conducted experiments on AIR-SARShip-1.0 in the case that the training samples with target-level annotations account for 10%–100% of all training samples that contain ship targets. The experimental results are presented in Fig. 10. The RefineDet is our baseline, and it can only use the training samples with target-level annotations to train. When all training samples containing ship targets have target-level annotations, we can get the upper bound of RefineDet.

As shown in Fig. 10, the F1-scores of the proposed method and RefineDet gradually increase as the training samples with target-level annotations increase. And the proposed method performs better than RefineDet in all proportion values, especially when the proportion is 10% and 20%, and the performance gap between these two methods is huge. When the proportion is 30%, the performance of the proposed method has exceeded the upper bound of RefineDet. And when the proportion is 100%, the proposed method can achieve the highest detection performance. At this time, the F1-score of the proposed method is nearly 5% higher than the upper bound of RefineDet. According to the above analysis, we can conclude that the proposed method can significantly improve the detection performance in the case of limited target-level annotations by utilizing the scene-level annotations effectively.

2) Ablation Study: To analyze the influence of different components in the proposed method, we implemented a series of ablation experiments on AIR-SARShip-1.0. Table II shows the quantitative experimental results. Same as the experimental settings in Section IV-D, only 30% of training samples that contain ship targets have target-level annotations. Please note that the proposed hierarchical test process from scene to target can only work under the premise of adding the scene classification loss in the training process. Therefore, we can not add the hierarchical test process from scene to target when the scene classification loss is not added.

As shown in Table II, we can see that our baseline, i.e., when the scene classification loss, scene aggregation loss, and the hierarchical test process from scene to target are all not added, has the lowest precision, recall, and F1-score. Then, we investigate the impact of the scene classification loss and scene aggregation loss in the scene characteristic learning branch by successively adding them to the baseline. The results in Table II turn out that these two losses can significantly improve the detection performance by effectively utilizing the scene-level annotations. In addition, the results in Table II also turn out that the proposed hierarchical test process from scene to target can improve the overall detection performance,

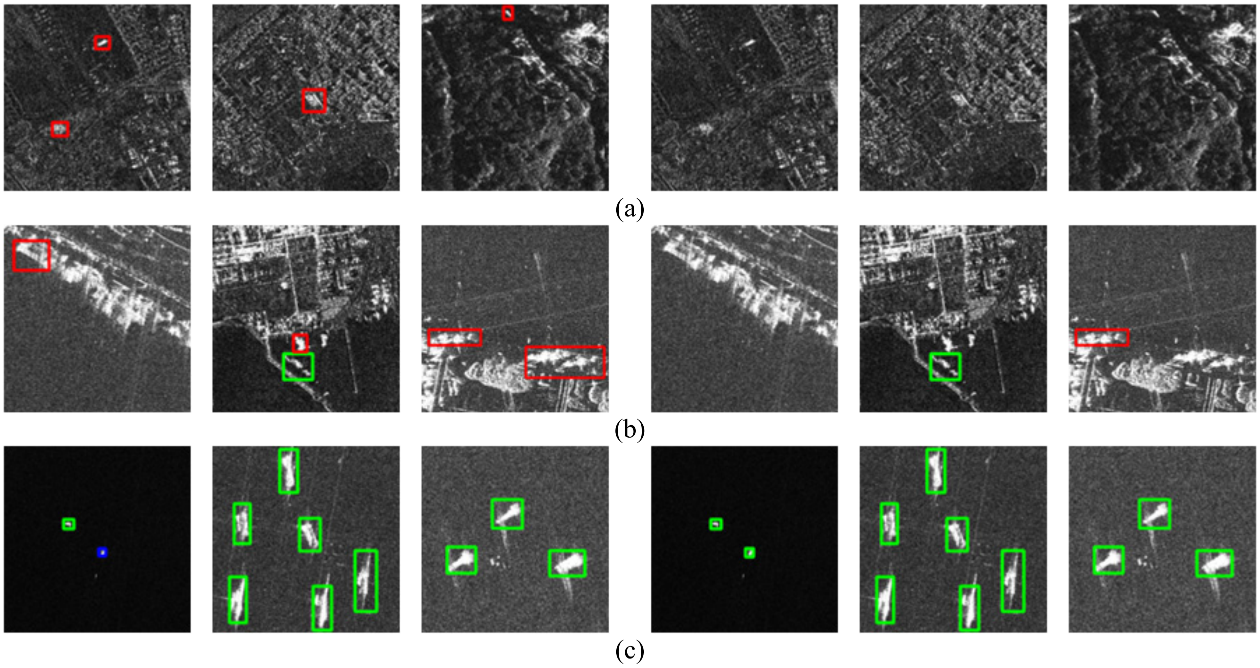


Fig. 11. Detection results of the traditional test process and the proposed test process on the inland, inshore, and offshore scenes, where the green rectangles represent the correctly detected target chips, red rectangles represent false alarms, and blue rectangles represent the missing alarms. Columns 1–3 are the detection results of the traditional test process, and columns 4–6 are the detection results of the proposed test process: (a) detection results on the inland scene; (b) detection results on the inshore scene; and (c) detection results on the offshore scene.

TABLE III
HYPERPARAMETER ANALYSIS EXPERIMENTS ON λ_1 AND λ_2 UNDER THE CONDITION THAT THE TRAINING SAMPLES WITH TARGET-LEVEL ANNOTATIONS ACCOUNT FOR 30% OF ALL TRAINING SAMPLES CONTAINING SHIP TARGETS

λ_2 \ F1-score \ λ_1	0.01	0.05	0.1	0.5	1.0
0.01	0.8167	0.8086	0.8184	0.8243	0.8335
0.05	0.8201	0.8387	0.8557	0.8469	0.8452
0.1	0.8308	0.8454	**0.8618**	0.8536	0.8477
0.5	0.8503	0.8447	0.8487	0.8610	0.8449
1.0	0.8317	0.8411	0.8501	0.8483	0.8438

especially can achieve higher precision, i.e., reduce the false alarms. To further verify the detection performance of the proposed test process on different scenes, Fig. 11 shows the detection results of the traditional test process and the proposed test process on the inland, inshore, and offshore scenes. Since the SAR images recognized as the inland scene did not participate in the target detection process, the proposed test process can significantly reduce inland false alarms, as shown in Fig. 11(a). From Fig. 11(b) and (c), we can see that the proposed test process can reduce inshore false alarms while reducing offshore missing alarms. It proves that setting different detection thresholds for inshore and offshore scenes is effective. At the end of Table II, we can see that the proposed method achieves the best detection performance when all components are added.

3) Influence of the Weight Factors λ_1 and λ_2 in Total Loss Function: In this section, to explore the influence of the weight factors λ_1 and λ_2, we changed the values of λ_1 and λ_2 and conducted experiments on AIR-SARShip-1.0. The experimental results are shown in Table III.

As shown in Table III, we observe that the proposed method can achieve the highest F1-score when λ_1 and λ_2 are both set to 0.1. The reasons may be as follows: the total loss of the network consists of the detection loss in the detection branch, the scene classification loss, and the scene aggregation loss. The weight factors λ_1 and λ_2 control the weight of scene

TABLE IV
PARAMETERS AND TOTAL TEST TIME OF DIFFERENT DETECTION METHODS

	Parameters	Total Test Time (Seconds)
SSD	2.37×10^{7}	21.74
RetinaNet	3.57×10^{7}	44.99
Faster R-CNN	1.36×10^{8}	140.51
Refinedet	4.29×10^{7}	21.16
YOLOX	2.53×10^{7}	23.76
Proposed method	4.41×10^{7}	16.63

classification loss and scene aggregation loss, respectively, in the total loss. When the values of λ_1 and λ_2 are set too small, the scene classification loss and the scene aggregation loss are of little help to the network, so that the feature extraction network has little improvement in the feature representation of ship targets and clutter. When the values of λ_1 and λ_2 are set relatively large, the network may not learn enough for the detection branch during the training process, thus affecting the detection performance. Of course, the optimal values of λ_1 and λ_2 may change for different datasets. However, the experimental results in this section can provide a rough reference for setting the values of λ_1 and λ_2.

4) Parameters and Total Test Time Analysis: In this section, we compare the parameters and total test time of the proposed method and other deep-learning-based detection methods. The experimental results on AIR-SARShip-1.0 are shown in Table IV. Since both STAC and Soft Teacher take Faster R-CNN as the baseline and their network structure is the same as Faster R-CNN, this section does not compare the STAC and Soft Teacher.

As shown in Table IV, we can see that the proposed method is significantly less than Faster R-CNN in terms of parameters but slightly higher than RefineDet. Although the SSD, RetinaNet, and YOLOX have fewer parameters than the proposed method, the total test time of the proposed method is the shortest. The reason is that the test SAR images recognized as the inland scene did not participate in the target detection process of the proposed method. In conclusion, the proposed method can not only significantly improve the detection performance but also improve the test efficiency.

V. CONCLUSION

In this article, we propose a novel semisupervised SAR ship detection network via scene characteristic learning. The proposed network constructs a scene characteristic learning branch. In this branch, a scene classification loss and a scene aggregation loss are designed to utilize the scene-level annotations. In this way, the scene characteristics of SAR images can be fully learned, thus enhancing the feature representation ability of the feature extraction network for ship targets and clutter. In addition, a hierarchical test process from scene to target is proposed. Under the premise of considering the scene characteristics of SAR images, we set different detection strategies for SAR images recognized as different scenes. The proposed test process can significantly reduce inland and inshore false alarms while reducing offshore missing alarms. The qualitative and quantitative experimental results based on two measured SAR datasets show that the proposed method can perform better than the other detection methods.

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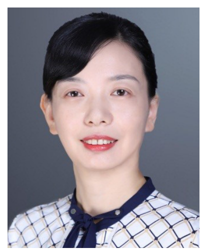

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